Qualitative and quantitative data

Hi again.

When it comes to decision-making, data is key.

But we've also learned that

there are a lot of different kinds

of questions that data might help us answer,

and these different questions

make different kinds of data.

There are two kinds of data that we'll talk about in

this video, quantitative and qualitative.

Quantitative data is all about

the specific and objective measures of numerical facts.

This can often be the what,

how many, and how often about a problem.

In other words, things you can measure,

like how many commuters

take the train to work every week.

As a financial analyst,

I work with a lot of quantitative data.

I love the certainty and accuracy of numbers.

On the other hand,

qualitative data describes

subjective or explanatory measures of

qualities and characteristics or

things that can't be measured with numerical data,

like your hair color.

Qualitative data is great

for helping us answer why questions.

For example, why people might like

a certain celebrity or snack food more than others.

With quantitative data, we can see numbers

visualized as charts or graphs.

Qualitative data can then give us

a more high-level understanding of

why the numbers are the way they are.

This is important because it helps

us add context to a problem.

As a data analyst,

you'll be using both

quantitative and qualitative analysis,

depending on your business task.

Reviews are a great example of this.

Think about a time you used reviews to decide

whether you wanted to buy something or go somewhere.

These reviews might have told you

how many people dislike that thing and why.

Businesses read these reviews too,

but they use the data in different ways.

Let's look at an example of a business using data from

customer reviews to see

qualitative and quantitative data in action.

Now, say a local ice cream shop has started using

their online reviews to engage with

their customers and build their brand.

These reviews give the ice cream shop

insights into their customers' experiences,

which they can use to inform their decision-making.

The owner notices that their rating has been going down.

He sees that lately his shop

has been receiving more negative reviews.

He wants to know why,

so he starts asking questions.

First are measurable questions.

How many negative reviews are there?

What's the average rating?

How many of these reviews use the same keywords?

These questions generate quantitative data,

numerical results that help

confirm their customers aren't satisfied.

This data might lead them to ask different questions.

Why are customers unsatisfied?

How can we improve their experience?

These are questions that lead to qualitative data.

After looking through the reviews,

the ice cream shop owner sees a pattern,

17 of negative reviews use

the word "frustrated." That's quantitative data.

Now we can start collecting qualitative data

by asking why this word is being repeated?

He finds that customers are

frustrated because the shop is

running out of popular flavors before the end of the day.

Knowing this, the ice cream shop can change its

weekly order to make sure it has

enough of what the customers want.

With both quantitative and qualitative data,

the ice cream shop owner was able to figure out

his customers were unhappy and understand why.

Having both types of data made it possible for

him to make the right changes and improve his business.

Now that you know the difference between

quantitative and qualitative data,

you know how to get different types of

data by asking different questions.

It's your job as a data detective to know

which questions to ask to find the right solution.

Then you can start thinking

about cool and creative ways to

help stakeholders better understand the data.

For example, interactive dashboards,

which we'll learn about soon.